Exam Instructions

2024-10-24. **Duration:** 3 hours. (14:00 - 17:00)

Start by reading all the questions in the 3 sections, to see if anything is unclear. If a question is unclear, please write down your assumptions about the question when you are answering it.

The exam has a total of 40 points, and you need 21 points to pass the exam (extra points will be added afterwards, once the exam is already graded). The grade is on the scale U/G.

Answers must be written in English. Dictionaries may be used.

For each question, I give the maximum points of the answer.
The format and extension of your answer should adapt to the type of questions:

Specific questions: a concept, a list... the *optimal* answer is usually around a single paragraph.

Short questions: explanations, reviews, comparisons... the *optimal* answer is usually a few (2 or 3) paragraphs.

Questions for reasoning: your arguments are presented in a concise, well-reasoned way, and written with your own words. The *optimal* answer to the whole question, meaning counting all sub questions, is usually around 2 pages or 3.000 characters. If the question asks you to provide an example, counter example, or justification:

Make sure it is original. An example or counter example is considered original if it has not been described in the slides of the course.

Make sure that is related with the scenario or application described at the beginning of the question.

<u>A checklist of common mistakes that cost points:</u>

Answer all **6** questions. A bad answer never gives less points than no answer.

- Read the question again after you have written the answer.
- Verify that you have answered the question.
- Verify that you answered all parts.
- Verify that you have not hidden the answer between many other irrelevant comments about the topic.
- *Cursives are added to highlight the key elements of the questions.*
- Don’t forget to give an example if that is requested and make it a concrete one.
- When a question asks you to compare two things A and B, make sure to highlight the contrasts: their differences. I do not want a full description of A and a full description of B, leaving it to me to find the differences. It is better to use tables than paragraphs of text to compare.

1 Question 1 [6 points]

1.a [02 points].

In the context of traditional Software Development Life Cycle (SDLC) models, describe two key differences between the Waterfall model and the Spiral model (01 point). Provide examples of scenarios where each model might be appropriately applied (01 point).

Fill in your answer here

Format

Words: 0

Maximum marks: 2

2 Question 1 [6 points]

1.b [02 points].

Discuss the importance of stakeholder management in software project management (01 point). Identify and explain two techniques that can be used to effectively manage stakeholders throughout a software project (01 point).

Fill in your answer here

Format

Words: 0

Maximum marks: 2

3 Question 1 [6 points]

1.c [02 points].

Explain the Bottom-up and Top-down cost estimation techniques (01 point). Provide an example of when each technique might be most appropriate in a software development project (01 point).

Fill in your answer here

Format

Words: 0

Maximum marks: 2

4 Question 2 [8 points]

2.a [02 points].

A software development team is using the Scrum framework. Differentiate the roles of the Scrum Master and Product Owner (01 point) and explain how they collaborate during a Sprint. (01 point)

Fill in your answer here

Format

Words: 0

Maximum marks: 2

5 Question 2 [8 points]

2.b [03 points].

Explain the importance of effective communication in project management (01 point). Describe three specific communication challenges that can arise in software development projects (01 point) and provide solutions for each challenge (01 point).

Fill in your answer here

Format

Words: 0

Maximum marks: 3

6 Question 2 [8 points]

2.c [03 points].

Explain the concept of requirements traceability (01 point) and describe two key benefits of implementing requirement traceability in a project (01 point) also explain how requirement traceability can assist in managing changes during the project lifecycle (01 point).

Fill in your answer here

Format

Words: 0

Maximum marks: 3

7 Question 3 [09 points]

3.a [03 points].

Differentiate the concept of cohesion and coupling in software design (01 points) and explain with example the importance of loose coupling in Service-oriented architecture (02 points).

Fill in your answer here

Format

Words: 0

Maximum marks: 3

8 Question 3 [09 points]

3.b [03 points].

Explain the concept of test-driven development (01 point) and discuss its advantages (01 point) and challenges (01 point).

Fill in your answer here

Format

Words: 0

Maximum marks: 3

9 Question 3 [09 points]

3.c [03 points].

Describe the V-Model in software development and testing (01 point). How does it relate to validation and verification activities? (01 point) Provide two advantages and two limitations of using this model (01 point).

Fill in your answer here

Format

Words: 0

Maximum marks: 3

10 Question 4 [06 points]

Question 4 [06 points]

Scenario: You are leading the development of a complex software system for a large financial institution. The system must be highly secure, reliable, and scalable to accommodate a growing user base and increased data volume. Your team is facing a critical decision: whether to adopt a monolithic architecture or a microservices architecture.

Questions:

a. Analyze the advantages and disadvantages of both monolithic and microservices architectures in the context of this scenario (02 points).

b. Based on your analysis, recommend the best architectural approach for this project, justifying your choice with specific reasons (02 points).

c. Explain how your chosen architecture will contribute to achieving the desired system attributes security, reliability, scalability and address potential challenges (02 points).

Fill in your answer here

Format

Words: 0

Maximum marks: 6

11 Question 5 [06 points]

Question 5 [06 points]

Scenario: A large software company is developing a critical healthcare system using an agile methodology. Midway through the project, the team realizes that certain security and compliance requirements were overlooked in the initial planning. The project is already behind schedule and over budget.

Questions:

a. Analyze the potential risks and consequences of this situation. (02 points)

b. Propose a strategy to address the overlooked requirements while minimizing impact on the project timeline and budget. (02 points)

c. Discuss how this situation could have been prevented and suggest three specific practices or techniques that could be implemented in future projects to avoid similar issues. (02 points)

Fill in your answer here

Format

Words: 0

Maximum marks: 6

12 Question 6 [05 points]

Question 6 [05 points]

Scenario: You are the project manager for a large-scale software development project. The project is currently behind schedule, and the client is requesting additional features. Describe how you would handle this situation using the following project management concepts.

a. Scope creep (01 point)
b. Agile methodologies (01 point)
c. Risk management (01 point)
d. Stakeholder communication (01 point)
e. Resource allocation (01 point)

Fill in your answer here

Format

Words: 0

Maximum marks: 5